

SUPERCONDUCTIVITY

INTRODUCTION

- **Superconductors**, materials that have no resistance to the flow of electricity, are one of the last great frontiers of scientific discovery.
- In 1911 superconductivity was first observed in mercury by Dutch physicist **HEIKE KAMERLINGH ONNES** of Leiden University, When he cooled it to the temperature of liquid helium, 4 degrees Kelvin (-452F, -269C), its resistance suddenly disappeared.
- In 1913, he won a Nobel Prize in physics for his research in this area.
- Super conducting materials have extra ordinary electrical and magnetic characteristics.
- Super Conducting materials have many important applications in the field of engineering & technology.

SUPER CONDUCTIVITY

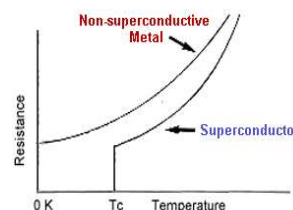
Definition: The phenomenon of sudden disappearance of electrical resistance in a material, when it is cooled below a certain temperature is known as **Super conductivity**.

- The temperature at which a material at normal conducting state changes into super conducting state is known as **transition temperature or critical temperature (T_C)**.
- T_C of a super conducting element is in the range of 0K to 9.5K. Examples: In-3.5 K He-4.2 K etc.

GENERAL PROPERTIES OF SUPER CONDUCTORS

a) Zero electrical resistance:

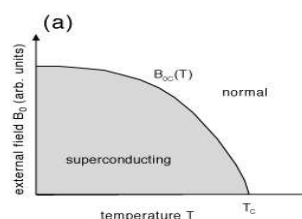
The electrical resistance of the superconductor is zero below the transition temperature (T_C). This property of Zero electrical resistance is known as ‘defining property’ of a super conductor.



b) Effect of Magnetic Field:

Below T_C , superconductivity can be destroyed by the application of a strong magnetic field. Minimum magnetic field strength required to destroy the super conducting property at any temperature is known as Critical magnetic field (H_C),

$$H_c = H_0 [1 - (T/T_C)^2]$$



where ' H_0 ' Critical magnetic field at 0K.

' T_C ' Super conducting transition temperature.

' T ' is the temperature below T

Figure: Application of magnetic field to a superconductor

c) Effect of electric current:

The application of very high electrical current to a superconducting material destroys its superconducting property.

Consider a wire made up of superconductor (i) be the current in wire, flow of high current induces a magnetic field, this induced magnetic field in the conductor destroys the superconducting property.

The critical current i_C required to destroy the superconducting property is given by,

$$i_C = 2\pi r H_C$$

where, ' H_C ' is the critical magnetic field, ' r ' is the radius of the superconducting wire.

- d) **Isotope effect:** The critical or transition temperature T_C of a superconductor is found to vary with its isotopic mass. i.e. *"the transition temperature is inversely proportional to the square root of isotopic mass of single superconductors."* This phenomenon is said to be isotope effect.

$$T_C \propto M^{-\frac{1}{2}} = T_c M^{\frac{1}{2}} = \text{const}$$

e) Effect of Pressure:

By applying high pressure, we can bring the T_C of a material nearer to room temperature, i.e., T_C is directly proportional to pressure.

f) Thermal properties:

Entropy and specific heat decreases at transition temperature. Thermo-electric effect disappears in the superconducting state.

g) Stress:

When a stress is applied to a superconducting material its dimensions vary.

h) Frequency:

At very high frequencies, the zero resistance of a superconductor is modified. The T_C is not affected by the frequency vibration.

i) Impurity :

The general properties especially the magnetic property of superconducting state are modified by the addition of impurities.

- j) **Size:** If the size of the specimen is reduced below 10^{-4}cm , the properties of superconducting state are modified.

MEISSNER EFFECT

The complete expulsion of lines of force by a superconducting material, when temperature of the specimen is below critical temperature when magnetic field is less than critical magnetic field is called "Meissner effect".

Consider a magnetic field is applied to a normal conducting material. The magnetic lines of force penetrate through the material. Consider that a normal conductor is cooled down to very low temperature for superconducting property. When temperature of the specimen is reduced to below critical temperature, then magnetic lines of forces are expelled by the

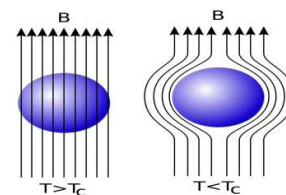


Figure: Meissner Effect

material. A diamagnetic material also repels the lines of forces. So, ejection of magnetic lines of forces, when superconducting material is cooled down is said to be the diamagnetic property. The diamagnetic property is explained using the following mathematical treatment.

The magnetic flux density is given by $B = \mu_0(M + H)$

For a superconducting material, $B=0$, Substituting $B=0$, we get

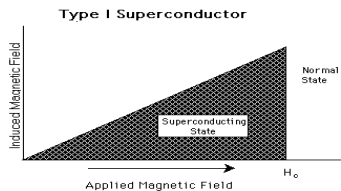
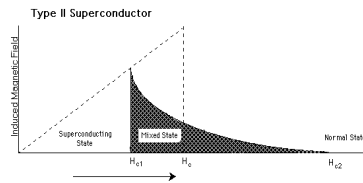
$$0 = \mu_0(M + H) \Rightarrow M = -H \Rightarrow \chi = -\frac{M}{H} = -1$$

The negative value of the susceptibility shows the diamagnetic properties of the superconducting material.

Based on Meissner effect, superconductors are classified into two types. They are

- i). Type-I superconductors
- ii). Type-II Superconductors

TYPES OF SUPERCONDUCTORS

TYPE-I Superconductors	TYPE-II Superconductors
- A Superconductor which exhibits perfect and complete Meissner effect is known as Type-I superconductors  Fig. 10 - Soft superconductors are those which can tolerate impurities without affecting the superconducting properties. - These superconductors are known as soft superconductors because they are easily magnetized or demagnetized. - Only one critical field exists for these superconductors. - The current flows through the surface only. - These materials have limited technical applications because of very low field strength value. - Examples are Pb, Hg, Zn, Al, Ga, W, In, Sn and Mo etc.	- A superconductor which doesn't exhibit perfect and complete Meissner effect is known as Type- II superconductor.  Fig. 11 - Hard superconductors are those which cannot tolerate impurities, i.e., the impurity affects the superconducting property. - These superconductors are known as hard superconductors because they are not easily magnetized or demagnetized. - It has two critical fields i.e H_{C1} (lower) & H_{C2}(upper), whose values are very high. - It is found that current flows throughout the material. - These materials have wider technology of very high field strength value. - Examples are Nb_3Ge, Nb_3Si, $NbTi$, NbN and V_3Sn

JOSEPHSON EFFECT

In the year 1962, Brian Josephson, an English Physicist discovered the tunneling of superconducting electron pair with opposite spin and momenta through a thin A(10-20Å) insulating barrier between two superconductors. This phenomenon is known as Josephson Effect.

- ♦ Two superconductors separated by a very thin strip of an insulator forms a Josephson junction.
- ♦ The wave nature of moving particles makes electrons to tunnel through the barrier. As a consequence of tunneling of electrons across the insulator there is net current across the junction. This is called d.c.josephson effect. The current flows even in absence of potential difference.
- ♦ The tunneling current 'I' is given by $I = I_0 \sin \Phi$
- ♦ The magnitude of current depends on the thickness of the insulators, the nature of the materials and the temperature.
- ♦ When a dc potential 'V' is applied across the junction of two superconductors which are separated by a thin insulating layer causes r.f. current oscillations across the junction. This effect is called a.c.josephson effect.
- ♦ The tunneling current 'I' is given by $I = I_0 \sin (\omega t + \Phi)$, where ω is angular frequency is given by $\nu = \frac{2eV}{h}$.

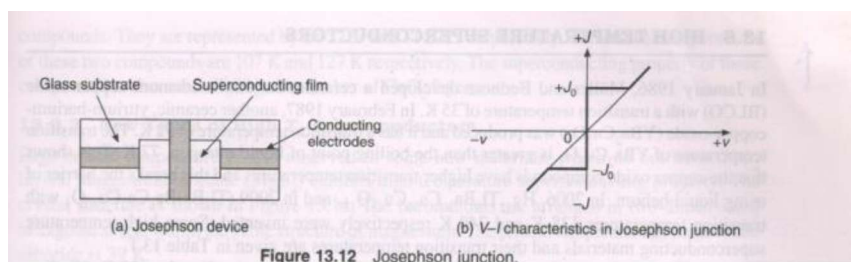


Figure: (a) Josephson device (b) I-V characteristics of Josephson junction

Characterstics of Josephson Junction:

- When applied volatge is zero, then the current flows across the junction is DC superconducting current and effect is D.CJosephson effect.
- When applied voltage is less than critical voltage, a constant dc current flows.
- When applied voltage is greater than critical voltage, then junction has finite resistance and current oscillates with radio frequency i.e $\nu = \frac{2eV}{h}$.

Applications of Josephson junction

A superconducting quantum interference device is a mechanism used to measure extremely weak fields, such as small changes in human body's, electromagnetic energy, using a device called a Josephson junction. These devices are used

- To detect magnetic field and it is used in magnetometer.
- Used in cardiology for Magnetic Field Imaging [MFI], this detects the magnetic field of the heart for diagnosis and risk stratification.
- Used to record the weak magnetic fields of the stomach as a magnetogastrography.
- A novel application is magnetic marker monitoring method, which is used to trace the path orally applied drugs.
- Used in magneto encephalography to make inferences about neural activity inside brain.

FLUX QUANTISATION

The magnetic flux that passes through a superconducting ring is quantized in the order of $\frac{h}{2e}$.

This phenomenon is said to be flux quantization. The value of the flux quantum

$$\varphi_0 = \frac{h}{2e} = 2.068 \times 10^{-15} \text{ Wb.}$$

Let us consider a superconducting ring to which magnetic field is applied in normal state, then lines of force penetrate through the material. When the temperature is reduced to less than critical temperature, then superconducting material expels the magnetic lines of force. When the magnetic flux is changed or removed, persistent current set up within the material remains same. This persistent current set ups due to magnetic flux which is equal to integral multiple of $\frac{h}{2e}$.

Magnetic flux with in ring $\varphi_0 = \frac{nh}{2e}$ where $n=1,2,3,4,5\text{-----}$

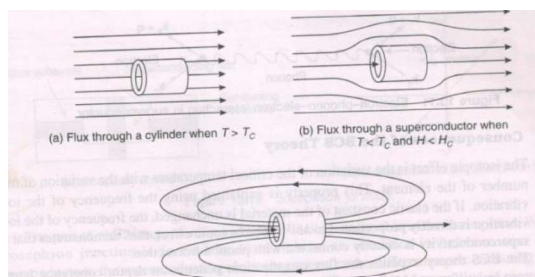


Figure: Quantization of magnetic flux in superconductors

PENETRATION DEPTH

Consider a superconducting material is subjected to a magnetic field. According to London, magnetic field does not drop to zero at the surface of the superconductors. It penetrates upto a small distance into the superconductors. The penetration of magnetic field with in the superconductor varies exponentially and is given by

$H = H_0 e^{-\frac{x}{\lambda}}$ where 'H₀' – magnetic field at the boundary of the specimen and 'λ' is called penetration depth

The **Penetration depth** is defined as the distance or depth in which the field decreases by a factor $\frac{1}{e}$. For a pure superconductor, penetration depth is in order of 500 Å. It is called penetration depth.

The penetration depth is also found to be temperature dependent. The temperature dependent of 'λ' is given by

$$\lambda = \lambda_0 \left[1 - \frac{T^4}{T_c^4} \right]^{-1} \quad \text{where '}\lambda_0\text{' is the penetration depth at 0 K and}$$

'λ' is the penetration depth at a temperature T K.

****BCS THEORY**

This theory is a quantum mechanical theory and it was proposed by Bardeen, Cooper and Schrieffer in 1957. Some of the important concepts of this theory was,

- i. According to this theory, two electrons with opposite spin and momentum pair up with one another due to a special type of attractive interaction. This pair of electrons is said to be **superconducting pair** or **cooper pair**. This pair get scattered, if there is sufficient energy so as to break up this pair of electrons. But in superconducting state, the energy is not sufficient to break up this pair and this pair moves undeviated through the impurities. This explains the low resistivity of the superconductor.
- ii. There is an attractive interaction between two electrons. This interaction is caused by the positive ions in the crystal. When an electron with wave vector (k₁) is passing closer to a positive ion, the electron transfers some of its energy to positive ion and moves with wave vector (k₁-q). Due to this there is a modification in the vibration of the positive ion. If another electron with wave vector (k₂) moves closer to this deformed positive ion, it interacts with the positive ion. The energy is transferred from positive ion to second electron as a result, its wave vector (k₁-q) changes. Hence a pair of electrons are formed through an attractive interaction.

$$k_1 - q = k_1' \text{ and } k_2 + q = k_2'$$

the resultant wave vector is $k_1 + k_2 = k_1' + k_2'$ and the wave vector is conserved.

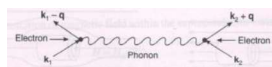


Figure: Electron-Phonon-electron interaction in superconductor

Consequences of BCS Theory:

- a. The BCS theory explains the flux quantisation, Meissner effect, penetration depth and zero resistivity.
- b. The energy gap of a superconducting material is given by $E_g = 3.5 K_B T_C$ at zero degree kelvin and becomes to zero at critical temperature.
- c. The de Broglie wavelength of the pair of electrons is infinitely long.

Applications of Super conductors:

- **Cryotron:-** Cryotron is a magnetically operated current switch.
- **Magnetic levitation:-** When a super conducting material shows the Meissner effect. Due to this effect, super conducting materials strongly repel external magnets. This leads to a levitation effect. When a magnet is placed over a super conductor, it floats. This phenomenon is known as Magnetic Levitation. Magnetic levitated Train(Maglev train).
- Superconductors are used as **Super conducting Magnets** able to produce **20 – 50 T**.
- Superconducting generators has the benefit of **small size** and **low energy consumption** than the conventional generators.
- **Very fast and accurate computers** can be constructed using superconductors and the power consumption is also very low.
- Superconductors can be used as **electric transmission lines** to transmit electrical power over very long distances without any **power or any voltage drop**.
- Superconductors are as **switching devices** such as relays.
- Superconducting devices are used for the **military applications** to detect the **nuclear submarines** passing through under water.